Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer and full-stack software developer with 9 years of production experience building large-scale systems and AI-powered applications. Built **AgentProbe** - an open-source multi-agent AI toolkit (LangGraph, FastAPI, ChromaDB, LLM-as-judge evaluation) live on GitHub. Delivered production AI across 1.6M+ financial records, HIPAA-compliant healthcare AI, and affordable housing compliance systems (Section 8/HCV, LIHTC, HUD). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 -- Aug 2025

Research Associate -- Fairfax, Virginia

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; designed Tableau, ggplot2, and Quarto dashboards improving faculty research comprehension by 40%.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed, improving downstream model reliability.

Entrata

Jan 2017 -- Aug 2024

Software Engineer

- Built 20+ full-stack microservices (Python, PostgreSQL, FastAPI, RabbitMQ) across 120+ Agile sprints for **100K+ monthly active users** -- 40% deployment time reduction; 99.5% uptime over 6+ months; cut execution time 50% via Docker containerization.
- Pioneered Entrata's AI engineering foundation -- built and internally shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Developed **affordable housing compliance modules** (Section 8/HCV, LIHTC, HUD) -- automated income eligibility verification, AMI threshold calculations, recertification workflows, and waitlist management across 100K+ unit portfolios.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; improved query performance 30% across 200+ SQL queries; led 300+ peer code reviews.

PROJECTS

AgentProbe: Multi-Agent AI Testing & Evaluation Toolkit

Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing -- deployed as a 15-endpoint FastAPI backend with CI/CD quality gates.
- Implemented **5-metric LLM-as-judge evaluation engine** (factual accuracy, hallucination, retrieval, completeness, compliance) blocking non-compliant agent runs before production; React dashboard tracking per-run metrics and regression drift.

PropLens: Lease Intelligence & Affordable Housing Compliance AI

In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs (Anthropic/OpenAI) · React

- Building RAG pipeline over lease PDFs -- LangGraph agents for compliance gap detection, LLM-as-judge scoring, and automated income eligibility verification for Section 8/LIHTC affordable housing across 1,000+ tenant records.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries (EN/ES) to FHIR healthcare APIs using LLMs, BERT, and semantic embeddings -- **HIPAA-compliant RBAC** across structured medical records.

CrashClarity: Large-Scale Road Accident Analysis

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages -- Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables -- deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Systems, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Prompt Engineering, Agent Observability, ChromaDB, Embeddings, OpenAI API, Anthropic API, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL, Oracle